The Transmission Efficiency of Net Investment Is Decreasing in the Replacement Burden

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Abstract

Standard analyses of investment and growth estimate a single transmission elasticity — one number summarizing how net investment maps into output growth. We show this number is not a constant but a function of the replacement burden $\delta = \text{CFC/VA}$, and that the function declines: a unit of net investment generates progressively less output growth as δ rises. We estimate the shape of this function — the gradient of the transmission coefficient across the δ distribution — on a primary country specification whose splitter and regressor carry no value added ($\delta_K = \text{CFC}/K$, net investment on the capital stock), so output enters only the outcome. Transmission attenuates by roughly 25–33% through the bulk of the distribution, significant under wild-cluster-bootstrap inference. A fully decoupled specification in which the splitter and regressor share no denominator at all yields the same monotone decline, and a placebo calibration shows the conventional output-denominator specification could reproduce the observed gradient only under transitory output-measurement noise an order of magnitude larger than national accounts admit. The same decline appears at three independent levels of aggregation — a 31-country OECD panel, a 9,019-observation OECD STAN industry panel, and a 251,130 firm-year Compustat panel — whose fixed-effects structures share nothing and whose observation counts differ by more than two orders of magnitude.

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1. Introduction

How much output growth does a unit of net investment generate? The question is usually answered with a single number — a transmission elasticity, estimated on a pooled sample and reported as a structural constant (De Long and Summers 1991; Mankiw, Romer, and Weil 1992). But as Acemoglu (2009, Ch. 1) cautions, the cross-country correlation between investment and growth does not by itself identify a causal effect; the level of any such elasticity is confounded by omitted factors common to investment and growth. This paper does not contest that. It shifts to a different object — the shape of the elasticity rather than its level — and shows that the elasticity is not a constant but a *function* of the replacement burden $\delta = \text{CFC/VA}$, the share of value added absorbed by replacing depreciating capital. The function declines: at high δ, net investment buys less growth than at low δ.

This reframing — from a transmission *coefficient* to a transmission *function of* δ — is the paper's first contribution, and it organizes a set of findings the macro-finance literature has treated separately. Tobin's q has risen while investment has fallen, and the share of capital that is short-lived has risen secularly.[1] These are symptoms of a weakening investment–growth link. We estimate the function whose decline the symptoms trace, and locate where along the δ distribution the decline sets in. The gradient also offers an account of the q–investment disconnect (Gutiérrez and Philippon 2017), developed in Section 7.

The second contribution is what makes the first credible: the decline replicates across three independent levels of aggregation. At the country level (31 OECD economies), the industry level (9,019 OECD STAN industry-country-years), and the firm level (251,130 Compustat firm-years), the transmission coefficient falls with δ. These panels use entirely different fixed-effects structures, are built from non-overlapping data, and differ in observation count by more than two orders of magnitude. A spurious gradient would have to arise independently, in the same direction, at all three.

The third contribution is methodological. The *level* of a transmission elasticity is not cleanly identified in observational data: net investment and growth share the output denominator, both respond to common shocks, and reverse causation runs from growth to investment. The *gradient* — a comparison of the coefficient in high-δ regimes to the same coefficient in low-δ regimes — differences these out, provided they do not themselves track δ. One does: because regimes are split on δ = CFC/VA and the regressor and outcome also carry value added, a transitory output shock moves all three together and is sorted by δ, so it does not net out of the difference. We treat this as the test the paper must pass, not a caveat to manage. Section 2 names the channel, and the country analysis answers it by stripping value added out of everything that defines or sorts the regime — re-estimating the gradient on capital-stock denominators — so that the channel is mechanically absent rather than assumed small. The decline is robust to that decoupling.

δ enters as a measured national-accounts ratio; the gradient is estimated as an empirical regularity, with structural interpretation deferred to Section 7.

The remainder proceeds as follows. Section 2 states what is claimed, names the one channel that threatens the gradient, and describes how the estimation removes it. Section 3 describes the data. Sections 4–6 estimate the gradient at the country, industry, and firm levels. Section 7 discusses.

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2. What Is Claimed, and the One Channel That Could Generate It

2.1 The object

Let g_{it} denote output growth in panel unit i in year t, n_{it} the corresponding net-investment rate, and δ_{it} the replacement burden. We estimate the transmission coefficient γ in

$$g_{it} = \gamma_{\text{above}}\, n_{it}\, I(\delta_{it} > \tau) + \gamma_{\text{below}}\, n_{it}\, I(\delta_{it} \le \tau) + \mu_i + \varepsilon_{it}, \qquad (1)$$

[1] Crouzet and Eberly (2023) re-attribute part of the q gap to intangible capital and rents; Corrado, Hulten, and Sichel (2009) and Haskel and Westlake (2017) document the secular rise of intangibles. These re-interpret the level of the gap; we estimate the gradient.

sweeping the cutoff τ across the δ distribution. The quantity of interest is the *gradient* of γ in τ — how transmission efficiency changes as one moves up the δ distribution — not the value of γ at any single τ.

2.2 The shape is identified where the level is not — and the one channel that survives the difference

The level of the net-investment-to-growth elasticity is confounded in observational data: n and g share the output denominator, both respond to common shocks, and reverse causation runs from growth to investment. Any single estimate of γ inherits all three.

The gradient is a *second difference* — the transmission coefficient in high-δ regimes minus the same coefficient in low-δ regimes — and a confound that biases the level of γ enters both regimes and cancels in their difference. This leaves a narrower class of threats: confounds that *themselves vary systematically with* δ. That class is not empty, and one member is specific and identifiable. Because the regime splitter is $\delta = \text{CFC/VA}$ and both n and g carry value added in their standard form (n as a rate over value added, g as value-added growth), a transitory shock to measured value added moves regime assignment, the regressor, and the outcome at once. The comovement it induces is sorted by δ by construction, so it does not difference away. We call this the *shared-output channel*, and it is the one channel we can name and remove.

Other candidates for δ-correlated confounding either work against the gradient or do not produce monotone decline. High-δ industries plausibly carry larger measurement error in the depreciation series — IPP deflators are noisier than structures deflators, and the gap between accounting and economic depreciation widens with the IPP share — but classical measurement error in the splitter attenuates the regime contrast toward zero rather than manufacturing it. Cyclical position is correlated with δ in the cross section (mature economies have both higher δ and lower trend growth), but cyclical comovement enters both regimes and would have to track δ within the same country-year cell to survive the difference; we have no mechanism by which it does so monotonically across the δ distribution. The shared-output channel is privileged because it is mechanically guaranteed to be δ-sorted by construction, and because it is the channel we can shut down at the design stage.

We do not bound this channel by assumption. We remove it. The cleanest move is to estimate the gradient on denominators that contain no value added, so that an output shock cannot reassign regimes or perturb the regressor — it can only add noise to the outcome, where it does not bias the slope. Concretely (Section 4), the regime is cut on $\delta_K = \text{CFC}/K$, net investment is measured on the capital stock ($n_K = \Delta K/K$), and only the outcome remains on the output side. In this specification the shared-output channel is mechanically absent, and we report it as the primary country result. The conventional output-denominator specification yields a similar point-estimate gradient, but its precision does not survive few-cluster-robust inference — which is consistent with part of that precision being the shared denominator at work, and is the reason we do not build the claim on it.

Two further checks fix the channel as a quantity rather than a possibility. First, a fully decoupled specification in which the splitter (δ_K) and the regressor (net investment *per worker*) share no denominator at all reproduces the decline (Appendix A, Table A1). Second, a calibrated placebo imposes a zero-gradient null, injects transitory multiplicative noise into the shared denominator, and asks how large that noise must be to fabricate the observed gradient (Appendix A, Figure A1): for the conventional output-denominator specification the answer is a transitory-noise standard deviation of roughly 4% of value added per year — about twice the largest credible national-accounts revision, and far below the level at which it would destroy identification rather than mimic it.

2.3 The shape we estimate

Across all three panels the transmission coefficient is monotonically lower in high-δ regimes than in low-δ regimes, and the attenuation is large. In the primary country specification the low-δ_K coefficient is near 0.55 and the high-δ_K coefficient is 0.35–0.41 across the 40th–70th percentiles of δ_K — an attenuation of 25–33%. The industry panel, which reaches further into high δ than the country panel, shows attenuation of 64–73% in the 18–25% range; a Hansen (1999) SSR grid search places the breakpoint at $\delta = 19\%$, within half a percentage point of the country-level break. The firm panel, whose denominators are not value added and so is structurally immune to the shared-output channel, shows attenuation rising from 48% at $\delta = 0.10$ to 86% at $\delta = 0.30$.

The three numbers agree on the shape — monotone decline from a stable low-δ transmission to a substantially attenuated high-δ transmission — even though the panels span more than two orders of magnitude in sample size and use entirely disjoint fixed-effects structures. The claim of this paper is the shape; Section 2.4 delimits where in the δ distribution it holds, and Section 2.5 delimits what the industry layer can carry.

2.4 The limit at the tail

We claim the onset of attenuation: transmission efficiency declines visibly through the low-to-middle range of observed δ, well before the distribution thins. We do not claim a terminal value in the high-δ tail, where the country panel turns sparse (fewer than 330 country-years above the 80th δ_K percentile) and the gradient loses statistical significance. The industry and firm panels extend much further into high δ and confirm the decline continues.

2.5 The limit at the industry layer

At the industry level we have not stripped within-industry temporal reverse causation: an industry that is both high-δ and slow-growing for unrelated reasons could contribute to the gradient through a dynamic channel we have not differenced away. The industry panel also has only eight country clusters and lacks the capital-stock and gross-investment series needed to repeat the decoupling, so it corroborates the *shape* across the wide-δ range but cannot itself address the shared-output channel; its significance should be read through wild-cluster-bootstrap inference, not naive clustered errors. The gradient is therefore a robust *cross-sectional and cross-level shape*, not a fully identified dynamic effect. The decoupling that answers the shared-output channel is carried by the country panel; the firm panel, whose denominators are not value added, is immune to the channel by construction (Section 6).

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3. Data

$\delta \equiv \mathrm{CFC/VA}$ is a standard national-accounts ratio (the consumption of fixed capital over value added); it requires no construction beyond what national and firm accounts report. We use it, and its value-added-free counterpart $\delta_K \equiv \mathrm{CFC}/K$, as measured covariates. The PWT depreciation series we use for δ_K is a model-derived asset-composition-weighted rate rather than a raw CFC/K ratio, so part of δ_K's cross-country and cross-time variation reflects shifts in asset composition (notably the rising IPP share); the shape we estimate is robust to whatever drives δ_K's cross-sectional variation.

Country panel. Two sources, used in parallel. The *value-added-free* panel — the basis of the primary specification — is Penn World Table 11.0 (Feenstra, Inklaar, and Timmer 2015) for 31 OECD economies, 1970–2023 (1,643 country-years): $\delta_K = \mathrm{CFC}/K$ from the PWT depreciation series, net investment on the real net capital stock ($\Delta K/K$ from `rnna`), and labour-productivity growth ($\Delta \ln(\mathrm{rgdpna}/\mathrm{emp})$) as the outcome. Output enters none of the quantities that define or sort the regime. The conventional *output-denominator* panel (OECD STAN 2025: δ, net investment rate, and value-added growth for 32 countries) is used only for the placebo calibration of Section 4. **Industry panel.** OECD STAN, 9,019 industry-country-year observations, δ from 0.2% to 106.5%, providing the wide-δ range. **Firm panel.** Compustat North America Annual, 251,130 firm-year observations on 23,955 firms, 1975–2023, excluding financials (SIC 6000–6999) and regulated utilities (4900–4999); firm $\delta = \mathrm{DP}/\mathrm{PPE}$, net investment rate $(\mathrm{CAPX} - \mathrm{DP})/\mathrm{PPENT}_{t-1}$, asset growth as the outcome; firm fixed effects, standard errors clustered by firm. At the firm level the denominators are property, plant, and equipment, not value added, so the shared-output channel of Section 2.2 does not arise by construction. Compustat DP is accounting depreciation and is not level-comparable to economic depreciation across layers; only the within-layer cross-sectional gradient is compared. All ratios are winsorized at the 1st and 99th percentiles.

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4. The Country-Level Gradient

The primary specification cuts the regime on $\delta_K = \mathrm{CFC}/K$ and measures net investment on the capital stock, so value added enters only the outcome and the shared-output channel of Section 2.2 is mechanically absent. The cutoff τ is swept across percentiles of the δ_K distribution (40th–90th) rather than absolute levels: cross-country differences in δ_K are small in absolute terms, and a percentile sweep gives a robust scan of the gradient that is comparable across the distribution. Inference is by wild cluster bootstrap (Rademacher weights, restricted null $\gamma_{\text{above}} = \gamma_{\text{below}}$, 399 replications), appropriate to the modest cluster count of 31.

Table 1: Country-level attenuation gradient, value-added-free denominators (PWT, LP growth). 31 countries, 1,643 country-years. Country FE; wild-cluster-bootstrap p on the gap $\gamma_{\text{below}} - \gamma_{\text{above}}$.

Cutoff (pctile of δ_K)	δ_K (%)	γ_{below}	γ_{above}	Attenuation	p (WCB)	N_{above}
40	3.47	0.554	0.413	25%	0.010	983
55	3.67	0.536	0.383	28%	0.003	740
70	3.92	0.522	0.351	33%	0.003	491
80	4.22	0.500	0.410	18%	0.160	329
90	4.59	0.494	0.423	14%	0.150	165

Net investment transmits to growth with a coefficient near 0.55 in the low-δ_K regime. Through the bulk of the distribution — the 40th to 70th percentiles of δ_K — the high-δ_K coefficient is 0.35–0.41, a 25–33% attenuation that is significant under few-cluster-robust inference ($p \leq 0.01$). In the sparse upper tail (the 80th percentile and above, fewer than 330 country-years) the gap narrows and loses significance; consistent with Section 2.4, we read the onset and the bulk, not the tail. Because the splitter, the regressor, and the regressor's denominator carry no value added, a transitory output shock cannot reassign regimes or move n_K; it enters only the left-hand side, where classical measurement error does not bias the slope. The shared-output channel cannot produce this gradient.

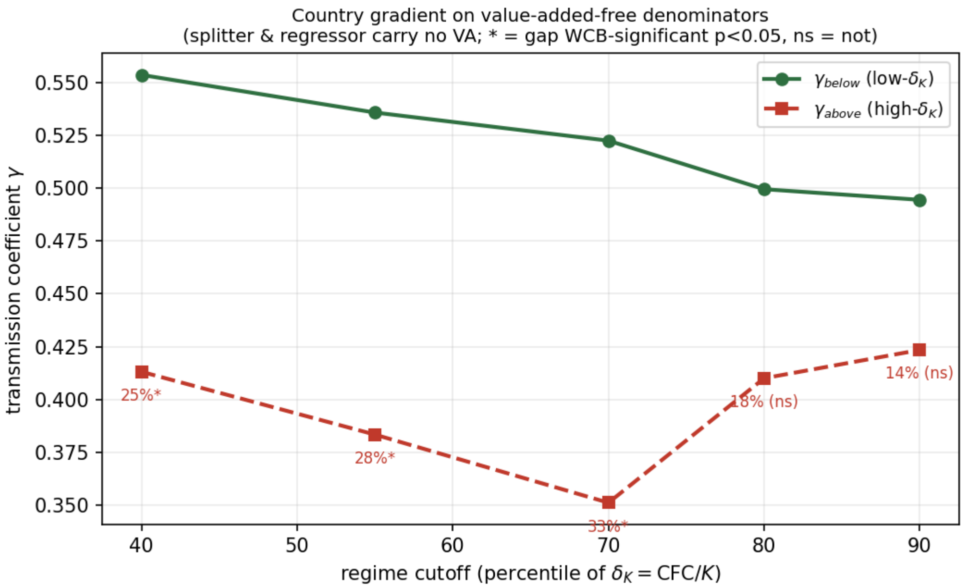

Figure 1: Country-level gradient on value-added-free denominators. The low-δ_K transmission coefficient γ_{below} is stable near 0.55; the high-δ_K coefficient γ_{above} lies well below it through the bulk of the distribution. Labels report attenuation; an asterisk marks gaps significant at $p < 0.05$ under the wild cluster bootstrap, "ns" those that are not.

Two checks confirm the channel is absent rather than merely small, and a third locates it as a quantity; full results are in Appendix A. (i) A *fully decoupled* specification replaces $n_K = \Delta K/K$ with net investment per worker ($\Delta K/\text{emp}$), so the splitter and regressor share no denominator whatever; the attenuation is 60–83% across cutoffs and significant at every one ($p \leq 0.015$, Table A1). (ii) A placebo calibration (Figure A1) shows that reproducing the conventional output-denominator gap under a zero-gradient null requires transitory value-added noise of roughly 4% per year, about twice any credible national-accounts revision. (iii) The same calibration applied to the residual shared-capital term in the decoupled specification requires about 2% — addressed directly by the fully decoupled specification, in which no denominator is shared.

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5. Industry-Level Replication

Table 2: Industry-level attenuation gradient (OECD STAN, LP growth). Country FE. The industry panel has eight country clusters; significance is read via wild-cluster-bootstrap rather than naive clustered errors.

τ (%)	γ_{above}	γ_{below}	Attenuation	N_{above}
12	0.112	0.313	64%	6,500
15	0.101	0.293	66%	5,239

τ (%)	γ_{above}	γ_{below}	Attenuation	N_{above}
18	0.081	0.303	73%	3,978
20	0.073	0.273	73%	3,362
25	0.071	0.211	66%	2,169
30	0.080	0.175	54%	1,380

The industry gradient runs from roughly 0.27–0.31 in low-δ industry-years to 0.07–0.09 in the 18–25% range. A Hansen (1999) SSR grid search places the threshold at $\delta = 19\%$, within half a percentage point of the country-level break. The value of this layer is its reach: the industry panel extends far into high δ, where the country panel is sparse, and the decline does not reverse. The role of this layer is correspondingly limited: with only eight country clusters and no capital-stock or gross-investment series, the industry data cannot repeat the decoupling of Section 4, so it confirms the shape over the wide-δ range the country panel cannot reach; it is not an identification layer. The decoupling that answers the shared-output channel is carried by the country panel.

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6. Firm-Level Replication

At the firm level the denominators are property, plant, and equipment, not value added, so the shared-output channel of Section 2.2 cannot operate: no value added enters the splitter or the regressor. This makes the firm panel the layer least exposed to the referee's concern, and the within-firm comparison the most direct instance of the second-order object of Section 2.1 — firm fixed effects absorb the level, leaving the shape. The 23,955 firms are US public firms; the firm panel therefore addresses the within-Compustat-universe cross-sectional shape, which is what the second-order claim requires. We sweep τ across absolute levels of δ (0.10 to 1.00) rather than percentiles because DP/PPE disperses widely at the firm level, and absolute thresholds anchor the regime contrast to economically interpretable depreciation rates rather than to distributional position.

Table: Firm-level attenuation gradient (Compustat, asset growth). 23,955 firms, 251,130 firm-years. Firm FE, SE clustered by firm. p is for $\gamma_{\text{above}} = 0$.[2]

δ threshold (DP/PPE)	γ_{below}	γ_{above}	Attenuation	p (γ_{above})	N_{above}
0.10	51.7	26.9	48%	< 0.001	196,910
0.20	52.1	15.9	69%	< 0.001	110,387
0.30	51.6	7.4	86%	< 0.001	71,429
0.50	50.4	0.1	99.8%	0.94	37,468
1.00	46.1	2.4	95%	0.03	14,816

The low-δ transmission coefficient is stable near 50 across thresholds, while the high-δ coefficient falls monotonically: attenuation rises from 48% at $\delta = 0.10$ to 86% at $\delta = 0.30$, each estimated with

[2] The gap $\gamma_{\text{below}} - \gamma_{\text{above}}$ is significant by orders of magnitude in this panel; we report the test on γ_{above} alone because the substantive question at the firm level is whether high-δ transmission has effectively vanished, not whether it differs from the low-δ benchmark.

high precision ($p < 0.001$) on tens of thousands of firm-years. Beyond $\delta \approx 0.30$ the high-δ coefficient is statistically indistinguishable from zero — transmission has effectively vanished — though the point estimates in the sparse high tail are correspondingly imprecise. High-δ firms convert net investment into asset growth at a small fraction of the low-δ rate. A supporting pattern: across the δ zones both mean growth and its variance fall through the transition; only in the sparse high-δ tail does variance rise, as outcomes detach from accumulation. We note it without leaning on the tail.

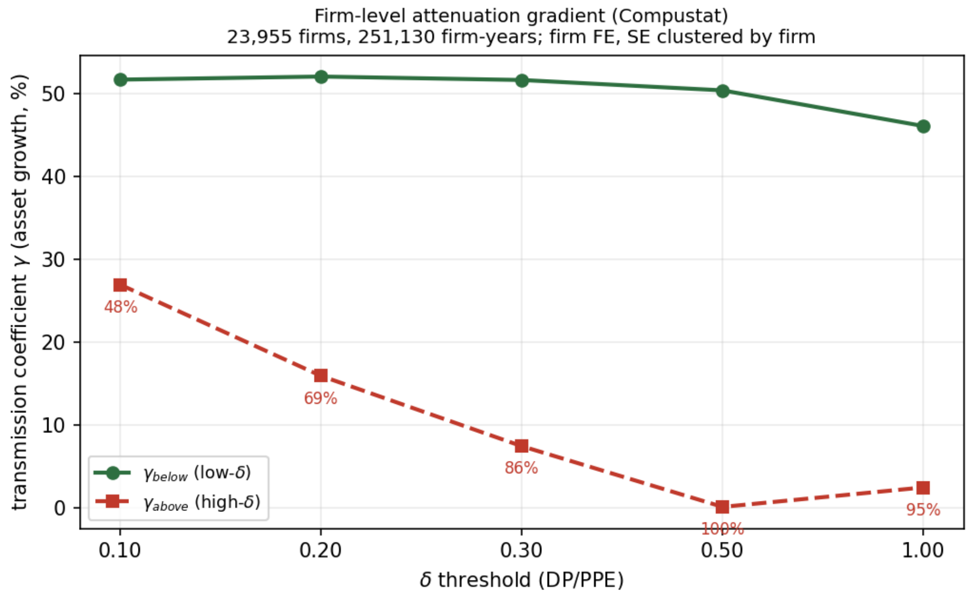

Figure 2: Firm-level attenuation gradient. The low-δ transmission coefficient γ_{below} is stable near 50; the high-δ coefficient γ_{above} declines monotonically and effectively vanishes by $\delta \approx 0.5$. Labels report attenuation at each threshold.

Because the firm splitter and regressor both carry property, plant, and equipment in their denominators, the firm analogue of the shared-output channel is a shared-PPE term. A placebo isolating it (Figure A2) shows it cannot generate the observed gradient: under a zero-gradient null, even transitory PPE noise of 40% per year manufactures a gap of about 6, against an observed gap of 44 — short by nearly an order of magnitude.

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7. Discussion

The gradient is a robust, cross-level *shape*: transmission efficiency declines with δ, visible from the low-to-middle range of observed δ and replicated at three independent levels of aggregation. The one channel that could generate it from no underlying relationship — the shared output denominator — is removed by construction in the primary country specification, which carries no value added in anything that defines or sorts the regime, and the decline survives. A fully decoupled specification, a placebo calibration, and a firm panel whose denominators are not value added all corroborate. The country decoupling carries the identification; the industry and firm panels carry the shape over a wide-δ range the country panel cannot reach. The three-layer alignment, with the country decoupling as its identified core, is what the claim rests on.

The gradient resolves the q–investment disconnect of Gutiérrez and Philippon (2017) as a δ-axis phenomenon. Tobin's q reflects the market value of accumulated productive structure — the marginal return to *gross* investment. But gross investment is increasingly absorbed by replacement as δ rises (Shi 2026a), and the marginal return to *net* investment — the residual that remains after replacement — declines along the δ gradient documented here. High-δ firms and economies therefore face high q and low net transmission simultaneously: q signals that the existing stock is valuable, while the gradient says that adding to it generates progressively less growth. The disconnect is not a governance or market-power failure; it is the financial-side projection of transmission attenuation along δ, paired with the real-side compression of net building.

A structural reading of the gradient is that the marginal return to net accumulation declines as the replacement burden rises. Why it declines — through diminishing returns at the accumulation margin or through rising coordination costs at the accumulation margin — is a question for structural work, which we leave to subsequent papers in this program.

Appendix A. Robustness of the Country Gradient

Fully decoupled specification. The regime is split on $\delta_K = \mathrm{CFC}/K$ and the regressor is net investment per worker ($\Delta K/\mathrm{emp}$), so the splitter and regressor share no denominator and neither the shared-output channel nor any residual shared-capital channel can operate. Coefficient levels depend on the per-worker units and are omitted; the attenuation ratio is scale-invariant.

Table A1: Fully decoupled country gradient (PWT). 31 countries, 1,612 country-years. Wild-cluster-bootstrap p on the gap.

Cutoff (pctile of δ_K)	δ_K (%)	Attenuation	p (WCB)	N_{above}
40	3.48	68%	0.003	961
55	3.68	70%	0.003	720
70	3.92	83%	0.003	487
80	4.23	60%	0.015	321

The gradient is significant at every cutoff, including the upper tail, and is larger than in the primary specification — consistent with the per-worker regressor carrying less attenuation bias from the shared denominator.

Placebo calibration. Under a zero-gradient null with constant transmission, transitory multiplicative noise is injected into the shared denominator and the regime regression is re-run; Figure A1 reports the manufactured gap $\gamma_{\text{below}} - \gamma_{\text{above}}$ as a function of the noise standard deviation.

Firm placebo. The firm analogue of the shared-output channel is the shared PPE between δ and n. Figure A2 injects transitory PPE noise under a zero-gradient null and finds that even a 40%-per-year standard deviation manufactures a gap roughly one-eighth of the observed gap at $\delta = 0.30$.

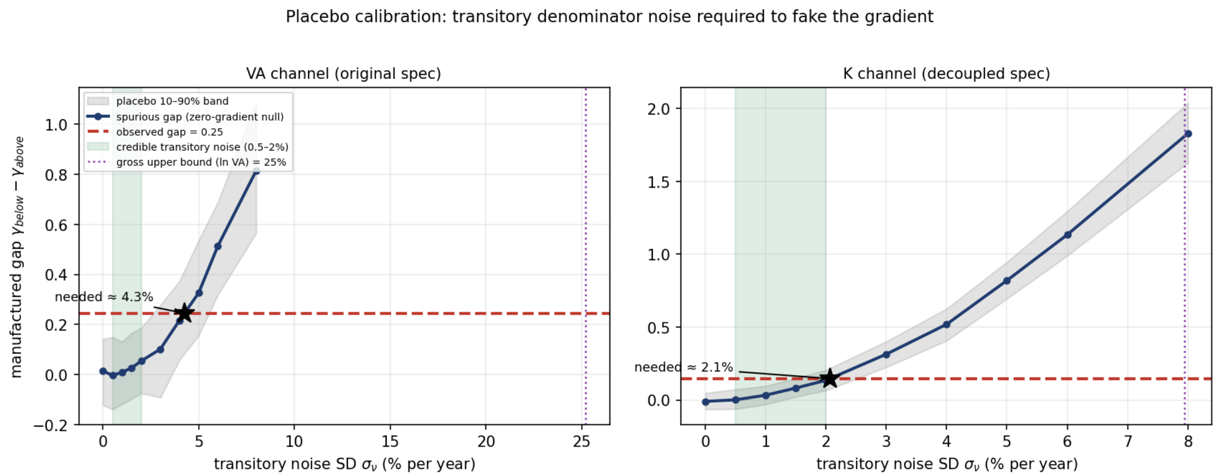

Figure A1: Placebo calibration for the country gradient. *Left (conventional output denominator):* reproducing the observed gap requires a transitory value-added-noise SD of about 4% per year — roughly twice the largest credible national-accounts revision (shaded band, 0.5–2%) and well below the gross upper bound given by residual $\ln \mathrm{VA}$ around a country trend (dotted line, 25%). *Right (capital denominator, decoupled specification):* the residual shared-capital channel requires about 2% — near the top of the credible band, against a much tighter gross upper bound (residual $\ln K$ around trend, 8%), reflecting that capital is a perpetual-inventory-smoothed stock; the fully decoupled specification above removes this channel entirely.

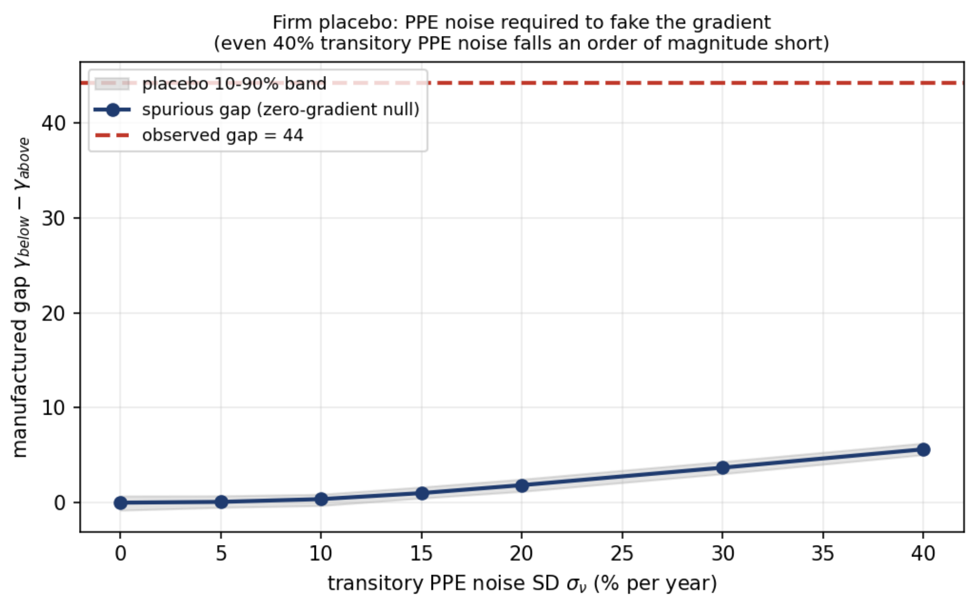

Figure A2: Firm-level placebo. Under a zero-gradient null, transitory PPE noise of as much as 40% per year produces a spurious gap of about 6 (with its 10–90% band), against an observed gap of 44 (dashed line). The shared-PPE channel cannot generate the firm gradient.

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